Frontier



Renual Inc.

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Renual Inc.

Organization location (we welcome applicants from anywhere in the world)

San Francisco, CA

Name(s) of primary point(s) of contact for this application

Laurent Stephane DEVRIENDT, PhD (CSO)

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit climate technology company deploying Enhanced Rock Weathering in the tropics and developing low-cost, field-portable MRV systems.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Field Validation of a Solid-Phase MRV System for Enhanced Rock Weathering

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Renual deploys Enhanced Rock Weathering (ERW) in the humid tropics and develops low-cost, high-accuracy measurement, reporting, and verification (MRV) technologies for soils in high-rainfall regions.

Enhanced Rock Weathering (ERW) is one of the most promising pathways for gigaton-scale carbon dioxide removal. Weathering of silicate dust on farmland generates alkalinity in soil and, where exported to the ocean, permanently removes 0.7–0.9 mol of atmospheric CO_2 per equivalent of alkalinity delivered (Hartmann et al., 2013). This process stores carbon in the ocean as dissolved inorganic carbon (DIC) for >10,000 years, i.e. the characteristic timescale for removal of excess oceanic alkalinity via $CaCO_3$ precipitation (Archer et al., 2009; Renforth and Henderson, 2017). In addition to CDR, ERW also delivers significant benefits to farmers: yield increases of 10–70%, reduced fertilizer dependence, and improved soil health (Jordan et al., 2025).

However, ERW's growth is constrained by the cost and complexity of measurement, reporting, and verification (MRV). ERW currently sits at Verification Confidence Level 2 (VCL2) in Frontier's framework, meaning every ton of removal carries a 40%+ uncertainty discount. This discount reflects the difficulty of measuring (1) how much Alkalinity was generated from rock weathering and (2) how much of this Alkalinity is exported to groundwater, rivers and the ocean.
The Isometric and Puro Earth registries currently require agreement between solid-phase and soil pore water MRV approaches to validate CDR credits. Achieving this agreement is extremely challenging due to large spatial and temporal variability in soil pore water chemistry. This challenge is particularly acute in tropical regions with high rainfall where soil water residence times are short and weathering products (Ca^{2+}, Mg^{2+}, HCO_3^-) are episodically flushed. Matching solid- and solution-based MRV approaches requires continuous monitoring of pore water chemistry at high spatial resolution, leading to prohibitive costs that are incompatible with scalable deployment. Moreover, a resolvable weathering signal in soil pore water may not emerge for the first several years of deployment due to alkalinity retention in various soil reservoirs (Bijma et al., 2026), creating a significant lag between feedstock dissolution and detectable alkalinity export.

Renual proposes to (1) field-validate a novel solid-phase MRV system that significantly reduces dependence on soil pore water sampling and (2) use the solid phase measurements to calibrate a reactive transport model for rice paddy and palm oil plantation soil profiles in Indonesia.

1. Our MRV system comprises two components: (a) a field-deployable device: the Integrated Weathering Capsule (IWC, patent pending) that allows field measurements of alkalinity released from feedstock dissolution; and (b) pre/post-application bulk soil sampling that quantifies total cation retention in soil (cf. Kanzaki et al., 2025). Together, these solid-phase measurements close

the soil cation budget (Fig. 1) entirely and allow accurate quantification of net alkalinity export from the soil layer, the key variable for CDR crediting:

$$TA_{export} = \sum (Cat)_{export}$$
$$= \sum (Cat)_{released} - \sum (Cat)_{retained}$$
$$= \sum (Cat)_{released} - (\Delta \sum (Cat)_{soil} - \sum (Cat)_{feedstock}),$$

where:

TA_{export} (in charge equivalents) is the total Alkalinity originating from feedstock dissolution that is exported away from the soil system ('Box 3' in Fig. 1).

$\sum (Cat)_{export}$ (in charge equivalents) is the sum of base cations ($2Ca^{2+} + 2Mg^{2+} + K^{+} + Na^{+}$) originating from feedstock dissolution that is exported away from the soil system ('Box 3' in Fig. 1).

$\sum (Cat)_{released}$ is the sum of base cations released from feedstock dissolution that is **directly measured with the IWC.** This variable accounts for the cation loss by secondary phase formation ('Box 2 – Box 8' in Fig. 1).

$\sum (Cat)_{retained}$ is the sum of base cations released from feedstock dissolution that are retained in the soil system (sum of 'Box 4 to 8' in Fig. 1).

$\Delta \sum (Cat)_{soil}$ is the change in the bulk soil cation pool **determined by pre- and post-deployment soil sampling**. This variable sums the retained cations and the undissolved feedstock fraction (sum of 'box 1' and 'Box 4 to 8' in Fig. 1).

$\sum (Cat)_{feedstock}$ is the sum of base cations remaining in the feedstock that is **directly measured with the IWC** ('Box 1' in Fig. 1).

In addition to the cation mass balance, changes in soil organic carbon (SOC) are also measured for full CDR accounting. ERW also boosts SOC accumulation over time via the formation of secondary phases with highly reactive surfaces. These secondary minerals form bonds with SOC, locking away organic carbon in soil for 100+ years (MAOM, Vienne et al., 2025; Fig. 1) .

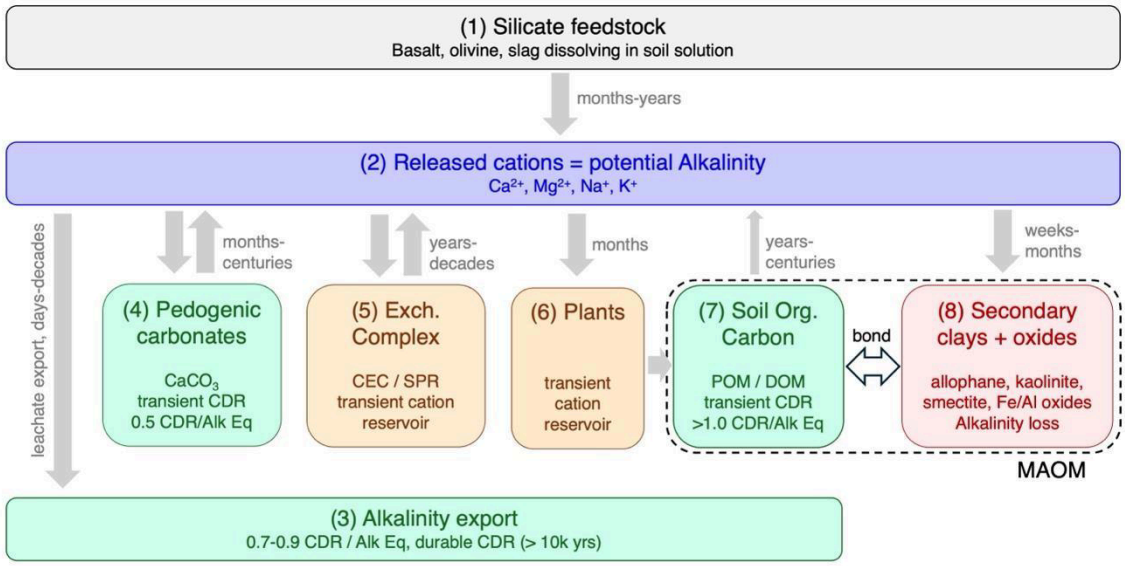


Fig. 1: Conceptual schematic of Alkalinity partitioning following silicate mineral dissolution in soil during enhanced rock weathering (ERW). Dissolution of silicate feedstocks (basalt, olivine, steel/Ni slag; gray, box 1) over months to years releases base cations (Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}; blue, box 2) that constitute the potential alkalinity pool. These cations partition among five competing sinks (box 3, 4, 5, 6 and 8) with distinct turnover timescales. Porewater export (box 3), pedogenic carbonates (box 4), soil organic carbon (box 7) are sinks that lead to CO_2 sequestration with different stoichiometric efficiencies and durability, expressed as moles CO_2 sequestered per equivalent of alkalinity generated (CDR / Alk Eq; green). The soil exchangeable complex (cation exchange capacity, CEC; surface protonation reactions, SPR) and biotic uptake by plants act as transient cation reservoirs that slow down durable CDR (orange/brown). The formation of secondary clay minerals and Fe/Al (oxyhydr)oxides is a quasi-permanent alkalinity loss (red) but these minerals significantly stabilize soil organic matter via the formation of mineral-associated organic matter (MAOM, dotted line).

2. CrunchFlow reactive transport modeling. We couple IWC field measurements with CrunchFlow (Steefel, 2015), 2017 R&D 100 award-winning mechanistic reactive transport software developed at Lawrence Berkeley National Laboratory (crunch.lbl.gov). CrunchFlow will be used to model the partitioning of feedstock-derived cations through key geochemical reservoirs: secondary phase precipitation, clay exchange sites, and pore water export (Fig. 1). Since 2022, CrunchFlow also includes a full calculation of the partially saturated water flow according to the Richard equation. The model will be calibrated using our solid-phase field data alone (IWC-derived dissolution rates and pre-/post-deployment bulk soil cation budgets) and then independently tested against soil pore water chemistry collected by Rhizon samplers. The pore water dataset is therefore not used in calibration; it serves as an out-of-sample benchmark for the calibrated model. Demonstrating agreement between predicted and observed pore water chemistry establishes that solid-phase measurements plus physics-based modeling can substitute for continuous aqueous monitoring at commercial scale.

Benefits to CDR community. We project MRV cost reductions of 70–90% relative to current liquid-phase methods, because: (i) 70–90% of our measurements are performed on-site using portable XRF with analysis costs <$5 per sample versus $30–80 for ICP-MS/OES; (ii) the IWC integrates the weathering signal over time, reducing the number of sampling events required; and (iii) the method eliminates the need for continuous pore water monitoring infrastructure. This cost structure is expected to bring the total all-in MRV cost below $15–25 per ton of CDR, enabling

profitable operation at CDR credit prices of $100-150/tCO$_2$, well within the range projected by Frontier and other advance market commitment buyers. CrunchFlow will be used to predict net alkalinity export based on soil type, climate parameters, and our low-cost in-situ weathering rate measurements from the IWC. The calibrated model will serve as a site-selection and CDR forecasting tool.

In addition to MRV cost reduction, the project will answer the following critical scientific questions:

(1) How is alkalinity released in soil partitioned between the different cation sinks illustrated in Fig. 1?
(2) How does alkalinity partitioning affect net CDR in the short (months to years) and longer time scales (decades to centuries)?

In other words, the project will resolve the characteristic timescales that determine what fraction of weathering-derived alkalinity is exported on climatically relevant timescales.

What needs to happen for step-change advancement. The IWC has been validated at bench scale. The critical gap is field validation with three main tests. First, the IWC must survive field deployment without mechanical or chemical compromise, preserving the integrity of the encapsulated feedstock and soil against the redox, biological, and hydrological stresses of tropical soils. Second, the IWC must remain in geochemical communication with its host environment so that internal weathering rates faithfully track in-situ conditions (pH, pCO_2, organic acid loadings, water saturation). Third, the bulk-soil leg of the MRV: quantifying retained base cations in soil requires a sufficiently low background in Ca, Mg, K, or Na (at least one) for the post-deployment enrichment to rise above pedogenic noise. Meeting these conditions would establish the first 100% solid-phase, pore-water-free MRV protocol for ERW.
Beyond these necessary conditions, a mechanistic reactive transport framework that reproduces CDR trajectories across geochemically distinct sites with a minimal, transferable parameter set, demonstrating that the platform generalizes beyond a single pedology rather than being tuned site-by-site.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Our project addresses three core research questions, each targeting a specific uncertainty component in ERW CDR quantification:

Research Question 1: Can the IWC accurately measure base cation release from feedstock dissolution in tropical field conditions?

The IWC is a PEEK-fabricated capsule in direct contact with native soil, deployed at root-zone depth. After 6–18 months of field exposure, the capsule is retrieved and the solid contents are analyzed by portable XRF to quantify elemental changes, specifically, the depletion of Mg, Ca and Fe (Na and K also possible but at lower precision) from the IWC matrix. The mass balance between

initial and final composition yields a direct measurement of total leached cations (i.e. not retained in secondary phases).

This approach has three advantages over conventional soil sampling: (i) the IWC eliminates spatial heterogeneity that plagues bulk soil measurements; (ii) the solid-phase signal integrates over the entire deployment period, removing sensitivity to episodic pore water flushing; (iii) portable XRF analysis is performed on-site in <5 minutes per sample at <$5/measurement, versus shipping to a centralized lab for ICP-MS at $30–80/sample with 2–4 week turnaround.

We will deploy 300+ IWC devices across 12 experimental plots (6 rice paddies in Java/Sulawesi, 6 oil palm plots in Sulawesi). Two feedstocks are tested: basalt (20 t/ha/yr) and nickel smelter slag (5 t/ha/yr), each with 3 replicates per crop type. Basalt is sourced from Java and/or Sulawesi volcanic provinces, Ni smelter slag from Sulawesi (~10 million tons/yr produced as industrial waste). IWC dissolution measurements will be validated against an independent method: pre- vs post-application bulk soil elemental composition analyzed via ICP-MS coupled with $^{87}Sr/^{86}Sr$ isotope source attribution to confirm that measured cation changes derive from the applied feedstock rather than background mineral weathering.

Research Question 2: What fraction of feedstock-released alkalinity is exported from the soil versus retained in secondary phases and exchange sites?

This is the central uncertainty in ERW CDR crediting. Feedstock dissolution releases alkalinity, but not all of it leaves the soil system. A fraction is retained in soil through three mechanisms: (i) precipitation as pedogenic carbonates (minimal in equatorial regions), (ii) uptake by secondary clay minerals and iron/aluminum oxides, and (iii) adsorption onto cation exchange sites (Fig. 1). The ratio of exported to retained alkalinity determines how much atmospheric CO_2 is permanently removed via the inorganic pathway.

We quantify alkalinity retention through paired measurements: IWC dissolution data (total alkalinity released from feedstock) combined with pre- and post-application bulk soil sampling that measures base cations in the upper 30 cm of the soil profile. The difference between released and retained alkalinity yields net alkalinity export. This mass balance closes entirely in the solid phase, no pore water sampling is required for the primary CDR calculation.

For independent validation, we will collect weekly to monthly soil pore water samples with Rhizon samplers and measure the dissolved cations via ICP-MS. This aqueous dataset plays two distinct, non-overlapping roles in our framework: (i) an independent check on the solid-phase cation budget for RQ2, and (ii) an out-of-sample benchmark for the CrunchFlow model in RQ3 (see below). It is never used to calibrate either the solid-phase budget or the model, a critical distinction from existing ERW MRV approaches that depend on aqueous data as the primary constraint.

Research Question 3: Can CrunchFlow reactive transport modeling predict alkalinity export from IWC + bulk soil measurements alone?

CrunchFlow (Steefel et al., 2015) is a multicomponent reactive transport code that simulates mineral dissolution, secondary phase precipitation, cation exchange, partially saturated flow and solute transport through porous media. We parameterize CrunchFlow with: (a) IWC-measured dissolution rates for each feedstock-soil combination, (b) bulk soil mineralogy and cation exchange capacity from pre-deployment characterization, (c) climate data (rainfall, temperature, evapotranspiration) from local meteorological stations, and (d) soil hydraulic properties from literature and field measurements.

The model predicts pore water chemistry and alkalinity export as a function of time and depth. The calibration step uses only solid-phase inputs (IWC dissolution rates, bulk soil cation budgets, mineralogy, and CEC); the model is then independently tested against the time series of soil pore water chemistry collected by Rhizon samplers. The pore water dataset is reserved as an out-of-sample benchmark and is never used in calibration. The key deliverable is a demonstrated correlation between CrunchFlow predictions (driven solely by solid-phase inputs) and observed aqueous-phase alkalinity export, with quantified prediction uncertainty from the three calibration cycles (6, 12, 18 months). If successful, this establishes that solid-phase measurements plus physics-based modeling can substitute for continuous aqueous monitoring, the step-change reduction in MRV cost and complexity that enables ERW scaling.

The calibrated CrunchFlow model will also serve as a site-selection and CDR forecasting tool. Given soil type, climate parameters, and feedstock composition, the model predicts expected CDR rates and alkalinity export efficiency for new deployment sites, reducing site-specific field trials and enabling rapid expansion of ERW across Southeast Asia and other tropical regions.

Location and scale. Field operations span two Indonesian sites and include two feedstocks (locally sourced basalt and Ni slag) selected to represent the dominant tropical ERW deployment contexts:

Java (rice paddies): Six 0.5 ha plots (3 ha) on lowland rice paddies near Jakarta, accessed through our MOU with Eratani (an agri-tech platform with ~34,000 smallholder farmers across Indonesia). We expect yield increases of 10–30% from liming effects and micronutrient release (Si, Ca, Mg), consistent with published ERW co-benefit data for tropical rice systems (Kantola et al., 2023; Jordan et al., 2025).

Sulawesi (oil palm): Six 0.5 ha plots (3 ha) on sustainable oil palm plantations in South Sulawesi, proximate to nickel smelting operations. The oil palm root system generates organic acids that accelerate dissolution, while the deep root zone (>1 m) creates distinct alkalinity transport dynamics compared to shallow-rooted rice.

Crop yield response will be tested across all 12 plots vs 6 control plots, generating the first systematic comparison of basalt and Ni smelter slag effects on rice and oil palm yields under identical tropical conditions. This yield data, combined with MRV cost data, will inform the farmer value proposition for ERW adoption at commercial scale.

Mesocosm experiments. In parallel with field deployment, we run controlled-environment soil column experiments at UC Davis (8 columns, 12–18 months, co-supervised by Prof. James Watkins) using the similar soil and feedstocks as with our Indonesian deployment. The columns are operated under accelerated weathering conditions, elevated temperature, defined pCO_2 and organic acid loadings, controlled water saturation, and tunable irrigation chemistry, designed to drive ≥10% feedstock dissolution within 6 months. The mesocosms serve two purposes. First, they provide an early stress-test of IWC mechanical and chemical integrity under accelerated weathering conditions, enabling design iteration on IWC.v2 before final field commitment and de-risking the 18-month field deployment. Second, they enable parallel testing of the full solid-phase MRV system (IWC + bulk soil mass balance) against high-resolution leachate ICP-MS, generating a fully closed cation budget under known boundary conditions, a definitive ground truth that is easier to attain than in heterogeneous field soils. The leachate dataset also provides an additional, independent benchmark for CrunchFlow under conditions where transport and boundary fluxes are explicitly controlled, complementing the field pore water tests of RQ3.

Differentiation from existing approaches. Current state-of-the-art ERW MRV relies on one of three approaches: (i) solid-phase isotope dilution requiring laboratory ICP-MS (Lithos, Silica), (ii) research-grade Field Monitoring Stations with lysimeters and rhizon samplers for continuous pore water monitoring (Inplanet, cost: $24.50/tCO$_2$), or (iii) periodic soil and water sampling with geochemical modeling (Mati, using SCEPTER/cGENIE). Each generation has improved on the last: Lithos pioneered cost-effective solid-phase attribution (2022), Inplanet demonstrated tropical soil-specific challenges (2022), Silica identified the Global South accessibility gap (2024). Renual integrates these lessons into a system that is solid-phase primary (eliminating aqueous solution dependency), field-portable (pXRF mostly replaces ICP-MS), and model-validated (CrunchFlow provides physics-based uncertainty reduction). CrunchFlow also improves on earlier geochemical modeling treatments by considering flow (if present) and solute transport.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

***IWC device longevity in tropical soils*.** The IWC is fabricated from PEEK (polyether ether ketone), among the most chemically and thermally resistant engineering polymers, and designed for multi-year deployment. However, we have not yet validated device integrity beyond bench-scale testing. Tropical conditions with high temperature (~25-30°C), intense microbial activity, root intrusion, and wetting-drying cycles may affect device integrity or introduce sampling artifacts. *Mitigation:* We deploy 324 IWC devices (27 per plot across 12 plots) and schedule retrieval of subset devices at gradually increasing intervals (week to multi-months) to assess degradation before full-term collection. If device failure occurs, we have backup designs that sacrifice some soil-feedstock contact for greater durability.

***CrunchFlow model fidelity for complex tropical soil profiles*.** CrunchFlow must reproduce the geochemical complexity of rice paddies and oil palm soils, including redox cycling (Fe^{2+}/Fe^{3+}), organic matter interactions, hydrological variability, and secondary mineral precipitation kinetics. The model may require iterative recalibration beyond initial parameterization. *Mitigation:* CrunchFlow already incorporates mineral dissolution by both strong and weak acids, cation exchange dynamics, secondary phase precipitation, and the major soil processes governing alkalinity retention (Steefel et al., 2015). Our project advisor Carl Steefel developed CrunchFlow and provides direct technical support for parameterization. We allocate three calibration cycles (6-, 12-and 18-month field data) to iteratively refine the model and quantify prediction uncertainty bounds.

***Feedstock trace metal management*.** Nickel smelter slag contains Cr, Ni, and Co at concentrations characteristic of ultramafic-derived feedstocks, comparable to the natural background of Sulawesi soils developed on ophiolitic parent material. At our 5 t/ha application rate, the incremental trace metal load represents a small fraction of native soil concentrations. We characterize each feedstock batch via TCLP-equivalent leaching tests and Cr speciation analysis prior to deployment, and we track soil and pore water trace metal concentrations at every sampling interval. Acceptance and continuation criteria are anchored to Indonesian PP 22/2021 soil quality standards and WHO guidelines, with documented procedures for batch rejection, plot-level discontinuation, and remediation should any threshold be approached.

***Project execution in Indonesia*.** Field operations require coordination across two islands, multiple local partners, and Indonesian regulatory requirements. *Mitigation:* Our MOU with Eratani provides established farmer relationships and local logistics. Laurent Stephane Devriendt (CSO) has conducted prior fieldwork in Indonesia. A collaboration with Bogor Agricultural University (IPB) provides continuous local presence and Indonesian government liaison.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1. Validated solid-phase MRV methodology (IWC + pXRF) with quantified accuracy, precision, and cost per measurement, benchmarked against independent aqueous-phase and isotopic validation 2. Calibrated CrunchFlow reactive transport model for rice paddy and oil palm soil profiles 3. Comprehensive field dataset: crop yield, dissolution rates, soil cation budgets, pore water chemistry, heavy metal monitoring across 2 feedstocks x 2 crop types x 3 replicates 4. Techno-economic analysis of MRV cost at commercial scale
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	Resolves the principal MRV bottleneck constraining ERW scaling. Reduces MRV cost by 70–90% (from \$25–50/tCO_2 to <\$5–15/tCO_2). Provides evidentiary basis to advance ERW from VCL2 (40%+ uncertainty discount) toward VCL3 (20%), effectively doubling saleable carbon removal. CrunchFlow model serves as transferable CDR forecasting tool for new geographies.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Q3 2026:** US: IWC manufacturing and initial US field testing. Indo Field trip planning. Outputs: 300+ IWC.v1 **Q4 2026:** US: Mesocosm experiments initiation at UC Davis. Implementation of tropical soil characteristics in CrunchFlow. Indo: Pre-deployment soil characterization. Start of IWC.v1 field deployment. Outputs: Adapted CrunchFlow model for rice and palm oil soil profiles **Q1 2027:**

	US: Potential production of IWC.v2 based on field testing results. Continued CrunchFlow and mesocosm works. Indo: First post-deployment large scale IWC.v1 field inspection. Outputs: IWC.v2? **Q2 2027:** US: 1st results from Mesocosm experiments (6 months). Indo: 1st post-deployment sampling and analyses (6 months) and CrunchFlow calibration. Deployment of IWC.v2? Outputs: 1st draft reports + TEA **Q3/Q4 2027:** US: 2nd results from Mesocosm experiments (12 months). Communication of preliminary results at AGU. Indo: 2nd post-deployment sampling and analyses (12 months) and CrunchFlow calibration. Outputs: 2nd draft reports + TEA **Q1/Q2 2028:** US: 3rd results from Mesocosm experiments (18 months). Complete dataset, 2-3 manuscript submissions and communication of results at international conference (e.g. Goldschmidt). Indo: Final post-deployment sampling and analyses (18 months) and CrunchFlow calibration. Outputs: Final reports + TEA, IWC commercialization
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	**US$709,000**